AR Network Architect

Interactive Augmented Reality Visualization of Network Topologies and Routing Algorithms

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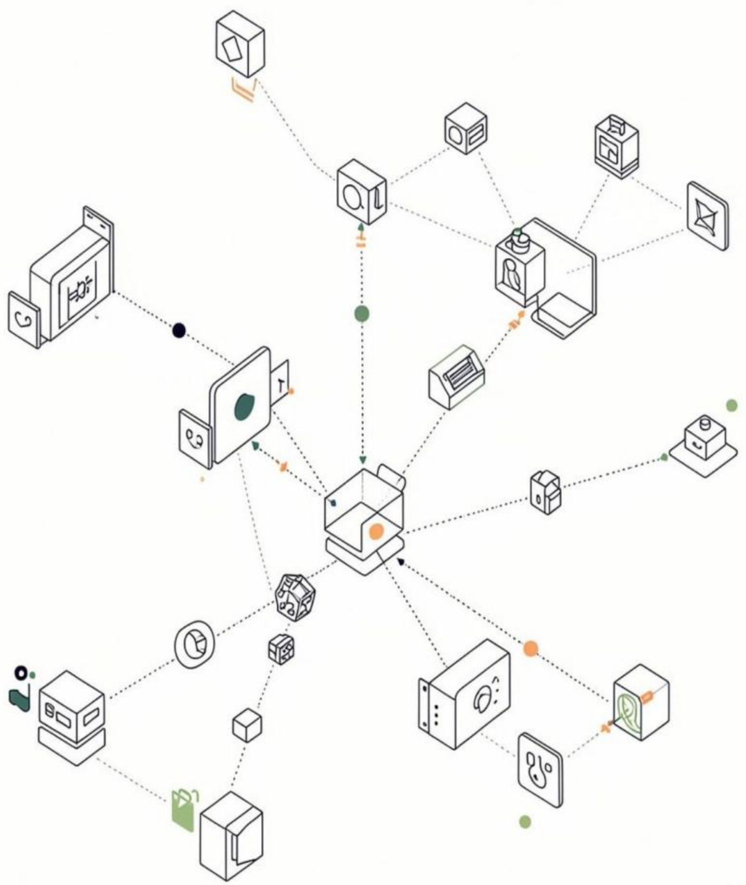


Why AR for Computer Networks?

Traditional Approach

- Static 2D topology diagrams
- Difficult to visualize dynamic packet movement
- Hard to understand spatial relationships
- Routing changes are mostly theoretical

Our Approach

- Place network nodes directly on a physical surface
- Build the topology interactively
- Visualize packet movement in 3D
- Observe routing changes immediately

Technologies Used

Frontend Layer

React.js + Tailwind CSS

AR & 3D Layer

WebXR Device API · Three.js / React Three Fiber · WebGL

Application Logic

Network Graph + Routing Algorithms

Deployment

Progressive Web App (PWA) · ARCore / ARKit compatible devices

System Architecture

How It All Connects

React handles application state and user interaction, while Three.js manages the 3D scene. WebXR bridges the virtual network with the physical environment through plane detection and pose tracking.

Each layer has a single responsibility — the architecture is modular and testable at every stage.

Placing Network Nodes in the Real World

Implementation Flow

01

Camera scans the physical surface

02

WebXR detects a horizontal plane

03

User taps on the detected surface

04

Ray cast returns intersection point

05

Network node placed at world position

WebXR Hit-Test Code

```
const hitTestResults =
 frame.getHitTestResults(hitTestSource);

if (hitTestResults.length > 0) {
 const pose = hitTestResults[0]
 .getPose(referenceSpace);
 placeNode(pose.transform.position);
}
```

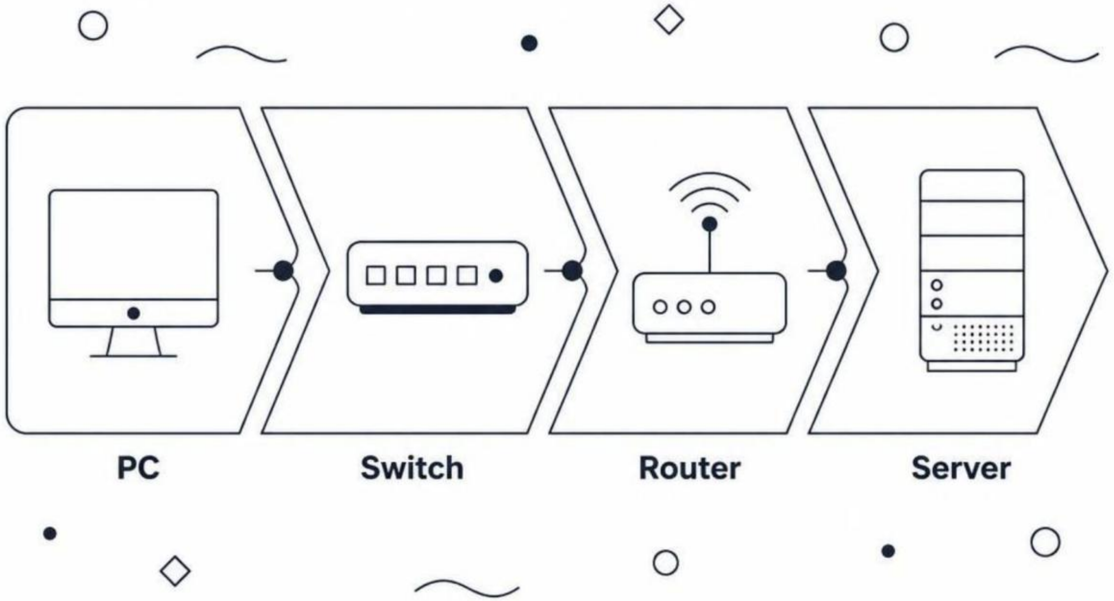
From Nodes to a Network Graph

Node Representation

- Each PC, switch, or router is a 3D object with a world position
- Connections stored as graph edges
- Same graph state drives both routing logic and 3D visualization

Node Data Structure

```
const node = {
  id: "router1",
  type: "router",
  position: [x, y, z]
};
nodes.push(node);
```



Calculating the Packet Route

Routing Process

1. Select source and destination nodes
2. Build network graph from placed nodes
3. Calculate edge weights using Euclidean distance
4. Compute shortest path and return node sequence

Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Routing Loop

```
for (const neighbour of graph[current]) {
 const distance =
 getDistance(current, neighbour);
 updateShortestPath(neighbour, distance);
}
```

Visualizing Data Movement

Packet Animation

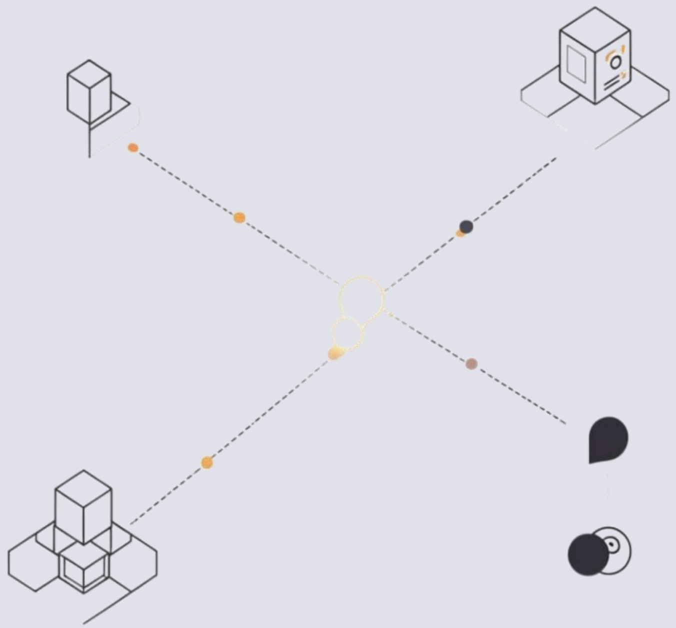
- 3D packet object follows the calculated route
- Smooth movement between nodes using linear interpolation (Lerp)
- Emissive / glowing material for visibility in AR
- Route is visually observable in the physical environment

Lerp Implementation

```
packet.position.lerp(targetPosition, speed);
```

Animation Path

1. Source Node
2. Intermediate Node 1
3. Intermediate Node 2
4. Destination Node



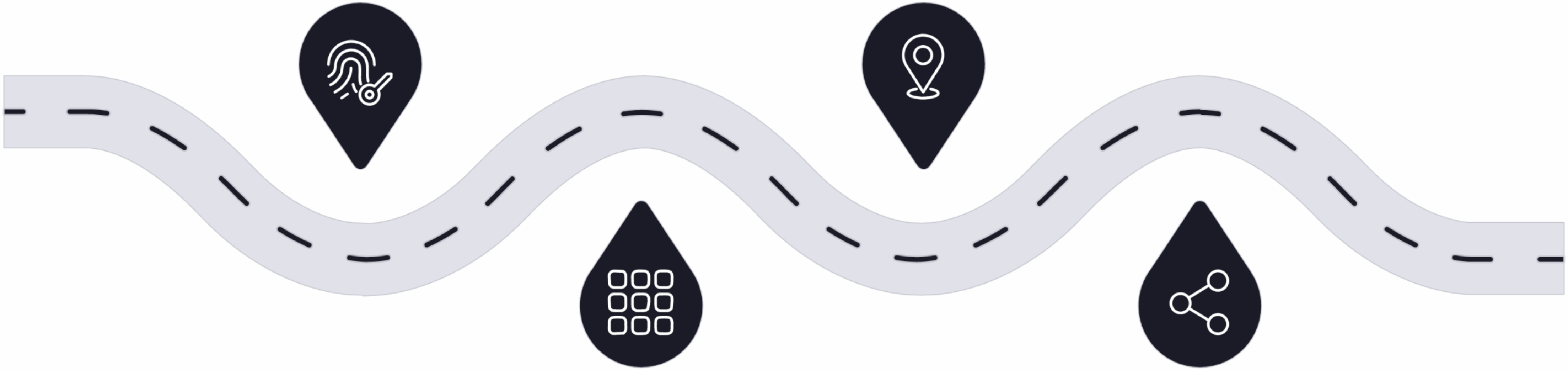
End-to-End Working

Scan

Capture the physical surface

Place Nodes

Position PC, switch, router, server



Detect Plane

Find a stable placement surface

Connect & Route

Link nodes and compute path

Example Scenario

PC → Switch → Router → Server

A packet is placed at the PC, routed through the switch and router, and delivered to the server — all visualized in AR on a physical desk.

If a connection is removed, the network graph updates and the route is recalculated in real-time — demonstrating dynamic topology changes.

What the Project Demonstrates

Implemented & Demonstrated

- Markerless AR-based network visualization
- Plane detection and hit-testing workflow
- 3D network node representation
- Network graph and distance-based routing
- Animated packet visualization
- React-based interface over AR scene

Project Scope

- Educational visualization tool
- PC, Switch, Router node types
- Packet route visualization
- Mobile AR interaction

Future Improvements

- Additional routing algorithms
- Packet inspection tools
- Real-time network traffic integration
- Multi-user synchronized AR

> *"The project converts an abstract network graph into an interactive spatial experience."*